FRANKLIN MBURU KAMANDE

Full Stack Software Engineer | Multi-Tenant & Enterprise Systems

Nairobi, Kenya
+254 718 921 677
franklinmburu05@gmail.com
GitHub: https://github.com/FranklinMburu
Portfolio: https://franklins-portfolio-v1.vercel.app/

PROFESSIONAL SUMMARY

Backend Engineer with 5+ years of professional experience delivering production-grade backend systems using Python (Django, Django REST Framework, FastAPI, Flask) and relational databases (PostgreSQL, MySQL). Proven track record designing secure REST APIs, multi-tenant architectures, and validation-heavy business systems where correctness, traceability, and reliability are non-negotiable.

Currently engineering a production multi-tenant enterprise CRM platform for a California-based company, working across tenant isolation, authentication/authorization, scalable data modeling, API reliability, and performance under real-world operational constraints. Experienced with Docker-based development/deployment workflows and cloud-hosted environments, collaborating with distributed teams through disciplined Git workflows, code reviews, and documentation standards.

Strong strengths in API security (JWT, RBAC), transactional integrity, audit logging, structured validation layers, and performance-conscious database design. Comfortable operating in structured, enterprise engineering environments and owning features end-to-end—from schema design and service-layer logic to API contracts and production readiness.

CORE TECHNICAL SKILLS

Backend & API Engineering

- Python (Django, Django REST Framework, FastAPI, Flask)
- RESTful API Design & Integration
- Multi-Tenant Architecture (Shared DB & Tenant-Aware Logic)
- Authentication & Authorization (JWT, RBAC, Session-Based Auth)
- Business Logic Enforcement & Validation Pipelines
- Error Handling, Logging & Monitoring Strategies
- Microservice-Oriented Design Concepts
- Background Tasks & Asynchronous Processing
- API Documentation (OpenAPI/Swagger)

Language & Stack Alignment

- Python (Primary Language – Production Use)
- JavaScript / Node.js (API Integration, Automation Workflows)

- Java (OOP, Backend Fundamentals)
- SQL (Advanced Querying & Optimization)

Cloud, Deployment & Runtime

- Docker (Development & Deployment)
- Cloud Hosting (AWS/GCP/Azure – deployment exposure)
- Linux Server Environments
- Production Configuration & Environment Management

Databases & Data Engineering

- PostgreSQL (Production)
- MySQL
- SQLite
- Schema Design & Normalization
- Indexing & Query Optimization
- Transaction Management & ACID Compliance
- Data Integrity & Audit Trail Systems

DevOps & Engineering Workflow

- Docker & Containerized Environments
- Linux Server Environments
- Git (Branching, Pull Requests, Code Reviews)
- CI/CD Concepts
- Environment Configuration & Secrets Management
- Postman & API Testing Tools

Systems & Enterprise Engineering Principles

- Clean Architecture & Separation of Concerns
- SOLID Principles
- Modular Codebases (Service Layer / Controller Layer Separation)
- Structured Logging & Traceability
- Secure Coding Practices (OWASP Awareness)

PROFESSIONAL EXPERIENCE

Software Engineer – Multi-Tenant Enterprise CRM

United Legal Group – Remote (California, USA)
Oct 2025 – Present

- Backend engineer on a production multi-tenant CRM serving multiple business clients.
- Designed and maintained Django REST APIs with PostgreSQL, enforcing strict tenant data isolation and secure access boundaries.
- Implemented and enforced role-based access control (RBAC) across tenant boundaries, including permission policies and validation rules.

- Built and maintained service-layer business logic and validation pipelines to enforce correctness for stateful workflows.
- Improved API reliability through consistent error-handling standards, structured logging, and defensive validation.
- Worked in Docker-based environments and contributed to cloud-deployed production configuration practices.
- Assisted with query optimization and performance improvements in high-usage modules.
- Collaborated in disciplined remote workflows: PR reviews, branching strategy, code quality checks, and documentation.

Impact: - Strengthened tenant-level security and access boundaries. - Improved onboarding clarity through documentation and environment standardization. - Reduced recurring configuration issues through structured setup guides and repeatable environments.

Senior Head of ICT

Stafford Schools – Nairobi, Kenya
Jan 2024 – 2025

- Led backend-heavy system development and digital transformation initiatives.
- Designed and deployed secure REST APIs using Django and FastAPI.
- Integrated financial transaction systems (including mobile payments) with callback verification and reconciliation workflows.
- Built structured reporting and validation pipelines reducing manual reconciliation errors by ~70%.
- Managed PostgreSQL and MySQL databases with performance tuning and schema optimization.
- Enforced code organization standards and production readiness practices.

Impact: - Reduced administrative workload by over 60% through automation. - Increased financial reporting accuracy through validation-driven workflows. - Improved system reliability via structured logging and transaction traceability.

SELECTED ENGINEERING PROJECTS

Multi-Layer Trading Execution Engine

Python • FastAPI • Modular Architecture • Risk & Validation Engine

- Designed a modular backend execution engine with clearly separated signal, validation, execution, and logging layers.
- Implemented deterministic rule-based logic with strict state validation to prevent invalid execution scenarios.
- Built configurable risk management constraints enforcing structured guardrails.
- Designed system for extensibility and production-readiness with structured logging.
- Applied test-driven validation principles to core logic modules.

Keywords: Backend Architecture, Risk Engine, Validation Logic, Structured Logging, Financial Workflows

Full-Stack Supplier Inventory Management System

Flask • React • PostgreSQL • JWT • RBAC

- Architected RESTful backend with modular controllers and service layers.
- Implemented JWT authentication and custom RBAC decorators.
- Designed normalized relational schema (Users, Roles, Suppliers, Customers, StockItems).
- Added audit logging for CRUD activity traceability.
- Implemented validation rules to enforce business constraints at API level.

Impact: - Improved inventory tracking accuracy. - Enabled secure multi-role access across operational users.

Keywords: REST API, RBAC, PostgreSQL, Backend Security, Audit Logging

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Movers Transport Management System

FastAPI • React • Mpesa API • SQLite

- Developed backend APIs coordinating vehicles, drivers, and assignments.
- Integrated Mpesa STK Push with secure callback validation and reconciliation logic.
- Automated assignment workflows reducing manual coordination by ~50%.
- Implemented structured validation preventing inconsistent operational states.

Keywords: Financial Transactions, API Integration, Workflow Automation, Backend Validation

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AI Reasoning & Memory Pipeline

Python • LangChain • ChromaDB • FastAPI

- Built RAG-based pipeline integrating vector database memory.
- Designed contextual retrieval workflows supporting state-aware reasoning.
- Structured modular execution layers separating retrieval, reasoning, and response handling.

Keywords: Data Processing, Backend Services, Vector Databases, Intelligent Systems

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EDUCATION

Software Development Diploma
Moringa School – Nairobi, Kenya
2024

- Backend Engineering
- REST API Design
- Database Systems
- Software Architecture Fundamentals

ADDITIONAL VALUE

- Strong alignment with enterprise engineering expectations: structured development, documentation, and correctness-first delivery.
- Experience working with U.S.-based remote teams (screen-sharing, async collaboration, and code review culture).
- Comfortable building in stacks commonly listed for backend roles (Python, Node.js exposure, Java fundamentals).
- Practical delivery mindset: secure APIs, data integrity, traceability, and production reliability.
- Able to own features end-to-end: schema design → services → API contracts → testing → deployment readiness.

AVAILABILITY

Open to Backend Developer / Software Engineer roles in structured engineering teams (Onsite or Remote). Available for full-time employment and long-term engagements.